Subject Name
FJFI ČVUT v Praze

Title of Work

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Date: **24.04.1945** Grade:

In the context of processing scientific papers, there are certain rules, some general, others differing depending on the educational institution. In this document, I will attempt to present all the rules that must be followed for your work to be correctly processed.

An example of processing a protocol chapter is given above. The protocol should consist of seven sections:

1 Work Tasks

Completely copied from your assignment. Individual points are marked with dots. If the assignment refers to a formula from the manual, the reference must be changed to point to the formula from your theoretical part. The same applies to images.

2 Equipment

Completely copied from your assignment.

3 Theoretical Introduction

MUST NOT be copied from the manual. It is necessary to briefly describe only the required theoretical information and derivation of formulas that you will use when processing the data.

All symbols in the formulas must be described in words in the text!

Example:

... for the electric current I the following law applies:

$$I = \frac{U}{R}, \tag{1}$$

where U is the voltage and R is the resistance.

Note that formula (1) is part of the sentence, so punctuation marks must follow it!

4 Measurement Procedure

This section describes in detail how the measurements were performed so that a reader can come to the laboratory and perform the experiment according to your procedure. It is also necessary to indicate how many times each quantity was measured.

5 Measurement Results

If your assignment included homework, in the first sentence of this section, write that the completed homework can be found in the Appendix in Fig. 3.

In this section, briefly but informatively describe what you measured and with what error, indicate where the measurement results are. Specify what is shown in the graphs; in the case of fitting, write the functions you used and the fitting constants you obtained.

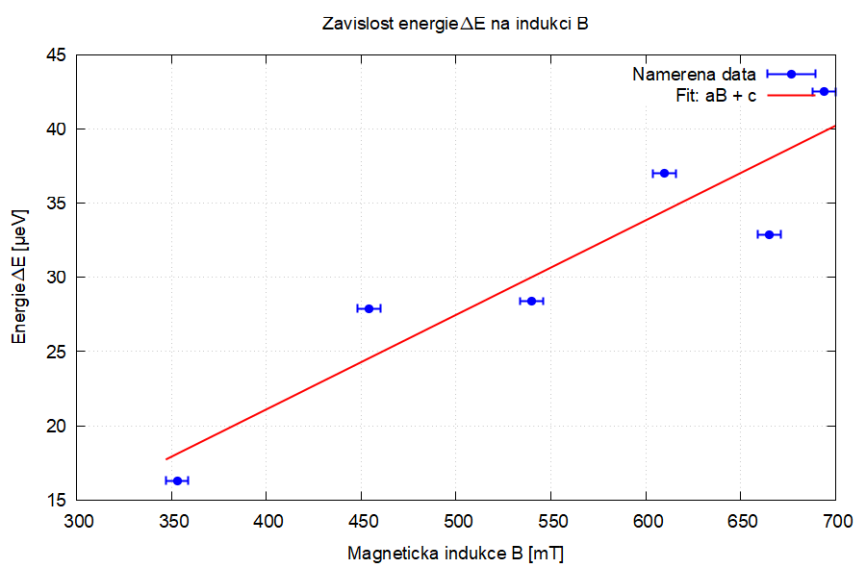
Example:

The measured values of radii $r_{\sigma-}$ with error $\sigma_{r-} = 0.05\,\mu\text{m}$ as a function of current I with error $\sigma_I = 0.1\,\text{A}$, and the calculated values of δ with error $\sigma_r = 0.3\,\mu\text{m}^2$, and the values of frequencies $\Delta\nu$ calculated according to equation (6) with the corresponding error $\sigma_{\Delta\nu}$ are given in Tab. 1. The errors were calculated as indirect measurement errors.

I [A]	$r_{\sigma-}$ [μm]	δ [μm^2]	$\Delta\nu$ [10^8 Hz]
2	2,3	0,8	36 $\pm$2
3	2,2	1,3	60 $\pm$2
4	2,1	1,5	72 $\pm$5
5	2,0	2,1	97 $\pm$6
6	2,0	2,4	113 $\pm$18
7	2,1	2,1	99 $\pm$9
8	1,9	3,0	140 $\pm$30

Tab. 1: Measured values of radii $r_{\sigma-}$ with error $\sigma_{r-} = 0,05\,\mu\text{m}$ as a function of current I with error $\sigma_I = 0,1\,\text{A}$, calculated values of δ with error $\sigma_r = 0,3\,\mu\text{m}^2$, and values of frequencies $\Delta\nu$ calculated according to equation (6) with the corresponding error $\sigma_{\Delta\nu}$.

Note that the text indicates which quantities were measured and which were calculated, and according to which formulas. It also shows how the error was calculated. Notice that the table description almost completely copies the written text. Pay attention to how measurement units are written in the table; a comma is used to separate the integer and decimal parts. If the error of a quantity is the same for the entire measurement column, it can be indicated only in the table description; if the error varies, it should be written in each row (see the last column).



Img. 1: Graph of the dependence of energy ΔE on the magnetic field strength B and fitting of the dependence with a function of the form $\Delta E(B) = aB + b$ with coefficients $a = 0,064 \pm 0,013$, $b = -4 \pm 7$.

Example:

The dependence of energy ΔE on B was plotted in the graph in Fig. 1, and by fitting with a function of the form $\Delta E(B) = aB + b$ we determined the values of the constants:

$$a = -0,064 \pm 0,013, \quad b = -4 \pm 7.$$

When describing the graph, do not forget to correctly label the axes and measurement units. For dimensionless quantities, write [-]. If you performed fitting, make sure to state the type of function and the obtained constants in the text and figure description (or directly in the legend).

6 Discussion

In this section, you discuss the obtained quantities and their errors. If you were supposed to experimentally verify a constant or a tabulated value, it is necessary to compare it with the tabulated value and decide whether these values agree within your measured error.

The procedure is as follows: take the tabulated value (for example from WolframAlpha), necessarily add the source to the bibliography and cite it, and check whether it falls within the 3σ interval, where σ is your error. In this case, the values can be considered consistent. Otherwise, you should discuss possible measurement issues and propose improvements.

To evaluate the quality of your measured quantity, you can also calculate the relative error of your measurement (divide σ by the measured value and multiply by 100 percent). If the quantity lies within the 3σ interval and has a small relative error, it means you have measured the desired quantity with only a small error. If the quantity lies within the 3σ interval but has a very large relative error, it is advisable to consider whether your measurement makes sense at all.

Example:

The manually calculated value of the Bohr magneton $\mu_B = (7,0 \pm 0,4) \cdot 10^{-5}$ eV/T has a relatively small error of about 6% and agrees within $\sigma \leq 3\sigma$ with the tabulated value $\mu_B = 5,788 \cdot 10^{-5}$ eV/T [2], therefore the values can be considered consistent.

However, despite the relatively small error, the calculated quantity is right at the 3σ boundary, which prompts us to consider possible sources of systematic error. One such source could be inaccurate readings on the scale in the eyepiece, as it was not always easy to set correctly.

The value of the Bohr magneton obtained using a CCD camera $\mu_B = (6,4 \pm 1,3) \cdot 10^{-5}$ eV/T has a relatively large error of about 20% and agrees within $\sigma \leq 3\sigma$ with the tabulated value $\mu_B = 5,788 \cdot 10^{-5}$ eV/T [2], thus the values can also be considered consistent.

In this case, we obtained a relatively large error of 20%. The main source of the error is fitting the obtained data. The problem lies in the relatively small number of measurements; the error could be reduced by performing more measurements.

7 Conclusion

The conclusion should indicate that you have completed all required tasks and state which quantities you obtained. You can simply copy the work tasks, replace words like "Measure quantity A"with "we measured quantity A,"and obligatorily state the value of this quantity A. If creating a graph was required, it is also necessary to include a reference to the graph. You should also mention the completion of homework and include a reference to the figure in the appendix.

Example:

In the homework preparation (see Appendix) we derived the interference condition $\Delta = k\lambda$.

We measured and determined the dependence of the magnetic field strength B between the poles of the electromagnet of the apparatus as a function of the current I through the coils (see Fig. 9) and manually determined the Bohr magneton $\mu_B = (7,0 \pm 0,4) \cdot 10^{-5}$ eV/T.

Reference

According to the rules, the first source in the bibliography should be the lab manual. References to all sources should appear at least once in the protocol text.

[1] KF Collective, *Manual: Exercise 9 – Light Polarization*, [online, cited March 13, 2011] - http://praktikum.fjfi.cvut.cz/pluginfile.php/415/mod_resource/content/test.pdf

[2] WolframAlpha: Bohr magneton [online, cited August 21, 2025]: https://www.wolframalpha.com/input?i=Bohr+magneton

Error Analysis

In this document, a detailed example is provided, which is to some extent suitable for all protocols you will encounter.

1 Measuring Instrument

When measuring any quantity with a measuring instrument, we must take into account the instrument's error. A simple rule applies: if the instrument is digital, take the unit in the last decimal place. If the instrument is analog (all instruments with a scale), take half of the smallest division.

instrument	type of instrument	measurement error
	digital	$\sigma_V = 0,1$ V
		$\sigma_W = 0,1$ W
	analog	$\sigma_l = 0,5$ mm $= 0,0005$ m

Sometimes the values on the instrument may fluctuate slightly; in such a case, it is necessary to increase the measurement error by a value of your choice.

2 Example

Assume we have a task of the following type:

Verify the law $A = kB$ and determine the value of the constant k. A and B are quantities we will measure.

Assume the quantity A is measured with a digital voltmeter. If the instrument displays four decimal places, we take the error as $\sigma_A = 0,0001$. The quantity B is measured with an analog instrument with error $\sigma_B = 0,05$ m.

As mentioned above, there may be a situation where the instrument shows a slightly oscillating value instead of a precise one. Assume our voltmeter shows variable values in the range approximately $1,0300$–$1,0700$ V on the first measurement. We then choose the value $1,05$ V and take the error not as the instrument error, but as $0,02$ V. In the protocol, it is necessary to state that we determined the error magnitude as $0,02$ V.

Rounding Rules

Which values should be recorded in the table? Assume that in determining the quantity F we obtained $F = 0,5689$ kg with error $\sigma_F = 0,036$ kg. These values must be properly rounded before further use (and when writing in the table). The rule is as follows: first round σ_F. Then round the quantity to the same order as the error:

$$\sigma_F = 0,036\,\text{kg} \rightarrow \sigma_F = 0,04\,\text{kg}; \tag{2}$$

$$F = 0,5689\,\text{kg} \rightarrow F = 0,57\,\text{kg}. \tag{3}$$

The final quantity is written as: $F = (0,57 \pm 0,04)$ kg.

These rules apply to all digits except 1. If the first significant digit of the error is 1, it is kept and the following digit is rounded according to rounding rules. For example, if we measured $F = 0,5689\,\mathrm{kg}$ with error $\sigma_F = 0,0167\,\mathrm{kg}$, the rounding is as follows:

$$\sigma_F = 0,0167\,\mathrm{kg} \rightarrow \sigma_F = 0,017\,\mathrm{kg}; \tag{4}$$

$$F = 0,5689\,\mathrm{kg} \rightarrow F = 0,569\,\mathrm{kg}. \tag{5}$$

The final quantity is written as: $F = (0,569 \pm 0,017)\,\mathrm{kg}$.

Assume that for each of the three values of quantity A, we measured the corresponding values of B, rounded them according to the rules above, and recorded them in Table 2 (first two columns). One possible procedure to calculate the value of the constant k is as follows: for each pair A and B, determine k from the formula $k = A/B$. The error of the quantity k is determined as the **error of indirect measurement**.

Assume we have a function $f(x_1, x_2, \ldots, x_n)$ of n variables. Let each variable x_i have an error σ_{x_i}. Then the error σ_f is determined by the formula:

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 \sigma_{x_1}^2 + \left(\frac{\partial f}{\partial x_2}\right)^2 \sigma_{x_2}^2 + \cdots + \left(\frac{\partial f}{\partial x_n}\right)^2 \sigma_{x_n}^2}. \tag{6}$$

In our case, the function f is given as $k = A/B$, where A and B represent x_1 and x_2. Then:

$$k = \frac{A}{B}; \tag{7}$$

$$\sigma_k = \sqrt{\left(\frac{\partial k}{\partial A}\right)^2 \sigma_A^2 + \left(\frac{\partial k}{\partial B}\right)^2 \sigma_B^2} = \sqrt{\left(\frac{1}{B}\right)^2 \sigma_A^2 + \left(-\frac{A}{B^2}\right)^2 \sigma_B^2}. \tag{8}$$

Measured values for each pair A, B are recorded in Table 2. In this way, we obtain three values of the constant k with the corresponding errors. To determine the final value of the constant k, we use the **weighted arithmetic mean**.

Assume we have n measurements of the same quantity x with **DIFFERENT** errors σ_{x_i}. Then the final value of the quantity x is determined as the **weighted arithmetic mean**, and the error as the error of the weighted arithmetic mean:

$$x = \frac{\frac{x_1}{\sigma_{x_1}^2} + \frac{x_2}{\sigma_{x_2}^2} + \cdots + \frac{x_n}{\sigma_{x_n}^2}}{\frac{1}{\sigma_{x_1}^2} + \frac{1}{\sigma_{x_2}^2} + \cdots + \frac{1}{\sigma_{x_n}^2}} = \frac{\sum_{i=1}^n \frac{x_i}{\sigma_{x_i}^2}}{\sum_{i=1}^n \frac{1}{\sigma_{x_i}^2}}; \tag{9}$$

$$\sigma_x = \sqrt{\frac{1}{\frac{1}{\sigma_{x_1}^2} + \frac{1}{\sigma_{x_2}^2} + \cdots + \frac{1}{\sigma_{x_n}^2}}} = \sqrt{\frac{1}{\sum_{i=1}^n \frac{1}{\sigma_{x_i}^2}}}. \tag{10}$$

If the errors σ_i are **EQUAL**, the above formulas can be simplified to obtain the formula for the **arithmetic mean** and the **error of the arithmetic mean**:

$$x = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}; \tag{11}$$

$$\sigma_x = \frac{\sigma_{x_i}}{\sqrt{n}}. \tag{12}$$

A [V]	B [m]	k [V/m]
$1{,}05 \pm 0,02$	0,8	$1{,}31 \ \pm 0,08$
$1{,}75 \pm 0,04$	1,3	$1{,}35 \ \pm 0,04$
$6{,}62 \pm 0,18$	5,0	$1{,}32 \ \pm 0,04$

Tab. 2: Measured values of voltage A with the corresponding error σ_A as a function of length B with error $\sigma_B = 0,05\,\mathrm{m}$, the values of the constant k calculated according to equation (7) with the corresponding error σ_k.

The errors σ_{k_i} are different, therefore:

$$k = \frac{\frac{k_1}{\sigma_{k_1}^2} + \frac{k_2}{\sigma_{k_2}^2} + \frac{k_3}{\sigma_{k_3}^2}}{\frac{1}{\sigma_{k_1}^2} + \frac{1}{\sigma_{k_2}^2} + \frac{1}{\sigma_{k_3}^2}} = \frac{\frac{1{,}33}{(0{,}08)^2} + \frac{1{,}35}{(0{,}04)^2} + \frac{1{,}32}{(0{,}04)^2}}{\frac{1}{(0{,}08)^2} + \frac{1}{(0{,}04)^2} + \frac{1}{(0{,}04)^2}} = 1,33444; \tag{13}$$

$$\sigma_x = \sqrt{\frac{1}{\frac{1}{(0{,}08)^2} + \frac{1}{(0{,}04)^2} + \frac{1}{(0{,}04)^2}}} = 0,02666. \tag{14}$$

After rounding, we can write the final value of the constant k:

$$k = (1,33 \pm 0,03)\,\mathrm{V/m}. \tag{15}$$

3 Determination of the Constant k Using the Fitting Method

Another way to determine the constant k is by **fitting** the measured data with a function. This method is used for more complex dependencies, especially when the function contains more than one constant.

To fit the data with a function, it is first necessary to roughly estimate the form of the function. In our case, it is sufficient to choose a function of the form $A = kB + C$, where C represents the shift of the graph along the y-axis.

The next step is a rough estimation of the constant values in this function (k and C are constants). We take the first measured values of A and B and calculate an approximate value $k \approx 1,3$. The value of C can initially be taken as zero, but it is better to avoid zero values, so we choose $C = 0,1$.

To fit the data, we use **Gnuplot**. The simplest code for this fit is:

```
f(x) = k*x + C            # Define the function
k = 1.3                   # Initial estimates of constants
C = 0.1

fit f(x) 'data.txt' using 1:2:3 via k,C   # Fit data from "data.txt"

set xlabel "B [m]"         # Set axis labels
set ylabel "A [V]"

set xrange [0.5:5.5]       # Set axis ranges
set yrange [0.5:7]
set key left               # Place legend on the left

plot 'data.txt' using 1:2:3 with yerrorbars title 'Data', \
     f(x) with lines title 'A = kB + C'   # Plot data and fitted function
```

The data for Gnuplot in the file `data.txt` in our case look as follows:

0.8	1.05	0.02
1.3	1.75	0.04
5.0	6.62	0.04

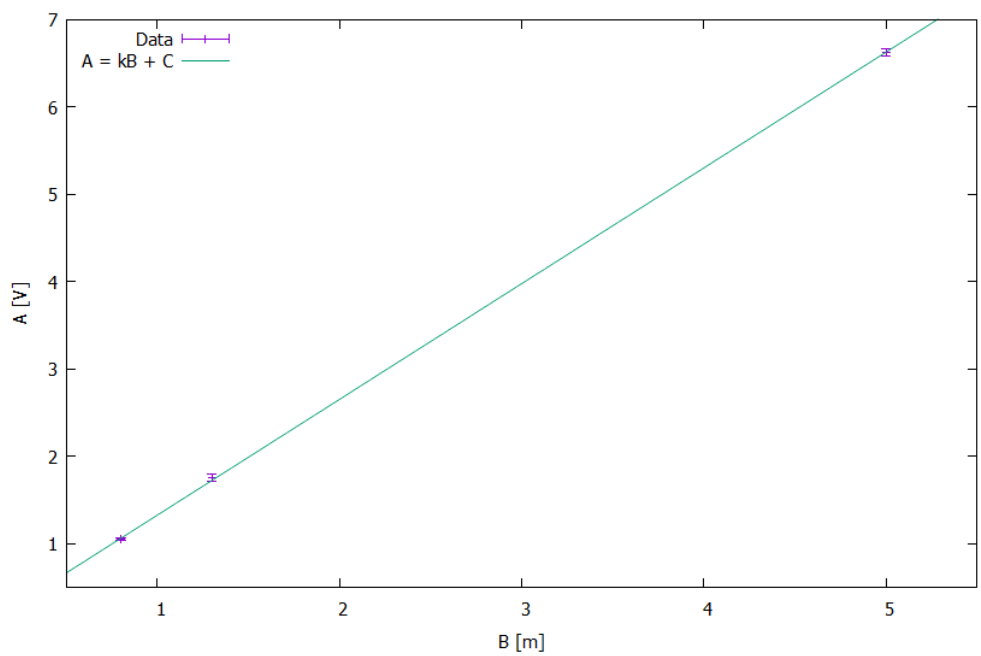
Note that the integer and decimal parts are separated by a dot. The first column corresponds to the values on the x-axis (B), the second to the values on the y-axis (A). The third column is the magnitude of the error σ_A. The line `using 1:2:3 via k, C` means using the values from the first column as the variable x, from the second as y, and from the third as the y-error. `via k, C` requests finding the values of the constants k and C. Unfortunately, Gnuplot cannot plot a function considering errors on both axes.

As a result of compiling the code, we obtain a graph, which is then inserted into the protocol (Fig. 2), and in the console we see the values of the fitted parameters C and k along with their errors:

```
Final set of parameters            Asymptotic Standard Error
=======================            =========================
k               = 1.32545          +/- 0.008907     (0.672%)
C               = -0.00363636      +/- 0.01967      (540.8%)
```

The constant C is not of interest to us, and the final value of the constant k after rounding is written as:

$$k = (1,325 \pm 0,009)\,\mathrm{V/m}. \tag{16}$$



Img. 2: Graph of the dependence of voltage A on the length B and fitting the dependence with a function of the form $A(B) = kB + C$ with coefficients $k = 1,325 \pm 0,009$, $C = 0 \pm 0,02$.